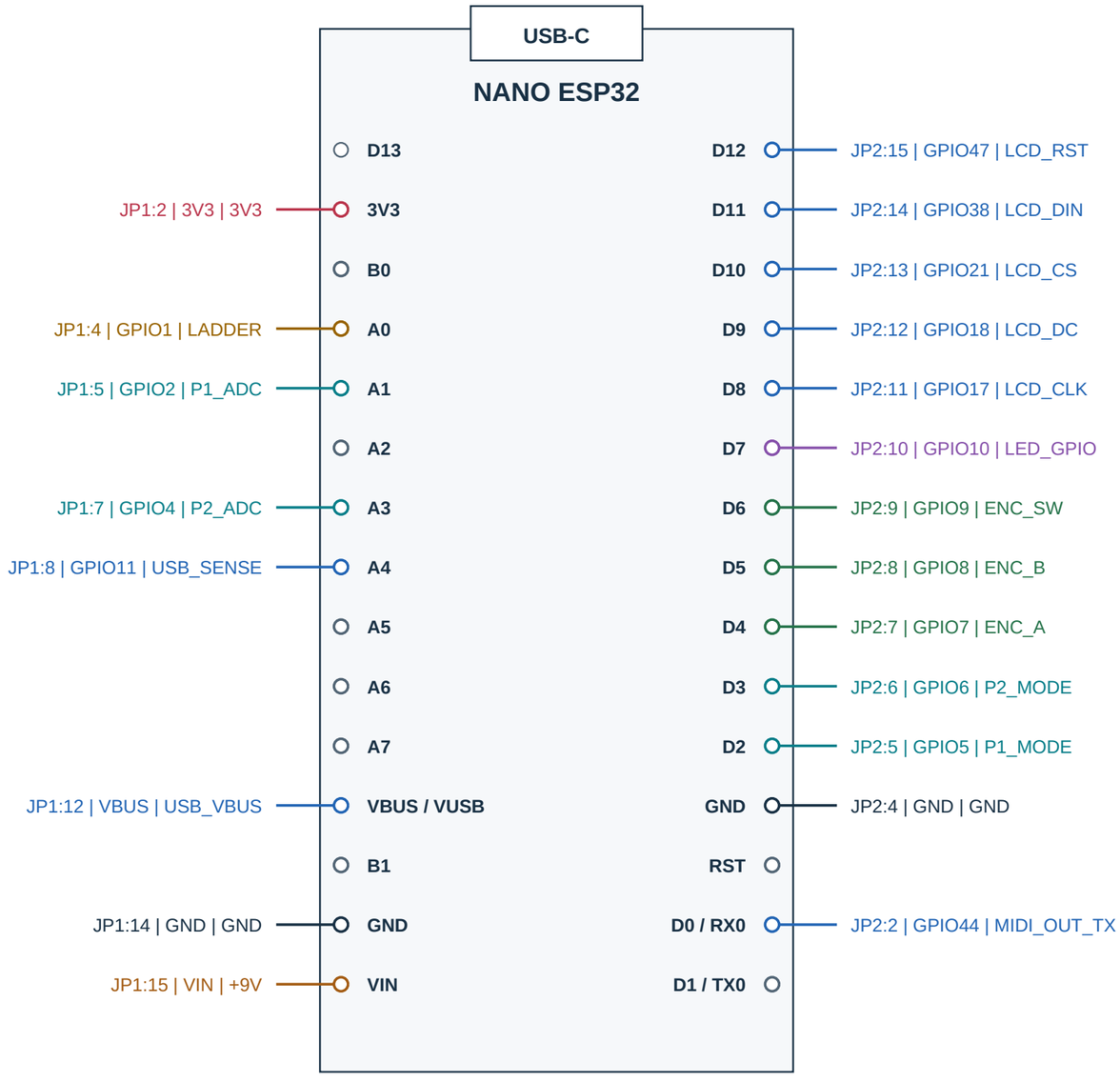
Controller and display - external pins

Arduino Nano ESP32 / ABX00083. Purchased boards stop at external pins; logical GPIO names accompany used board labels.

U1 Arduino Nano ESP32

Component-side top view, USB-C up. All printed labels retained; unused outside callouts omitted.



JP1: 1-15 top-down; JP2: 15-1 top-down (official schematic pad numbering).

VBUS is labelled VUSB in the official pinout. RST is RESET in that pinout.

Native USB MIDI uses the USB-C connector. B1 / GND + RST enters the ROM flasher.

U2 Waveshare 24382

P2 - 8-pin GH1.25 connector

Pin	LCD label	Nano label / GPIO
1	BL	3V3
2	RST	D12 / GPIO47
3	D/C	D9 / GPIO18
4	CS	D10 / GPIO21
5	SCL / CLK	D8 / GPIO17
6	SDA / DIN	D11 / GPIO38
7	GND	GND
8	VCC	3V3

SCL and SDA are SPI signals here.

No MISO connection is required.

BL ties to VCC (3V3) for always-on backlight.

Use the external P2, not the LCD FPC.

Display orientation

Landscape 280 x 240 pixels.

Left variant: firmware rotation 180 degrees.

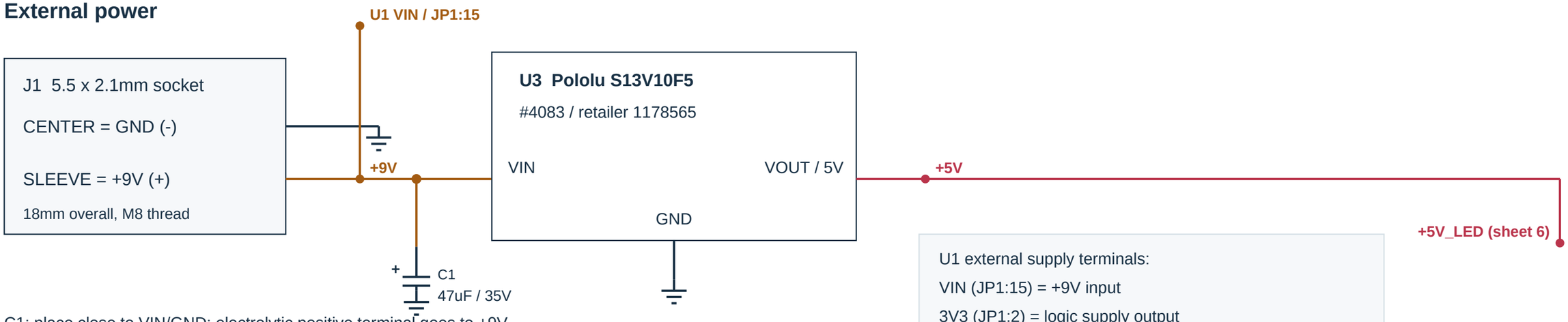
Right variant: firmware rotation 0 degrees.

The electrical wiring stays identical.

Power distribution and rotary encoder

9V center-negative input; 3V3 logic; VIN receives 9V; converter 5V powers only optional LEDs. USB VBUS is sensed, not joined to 5V.

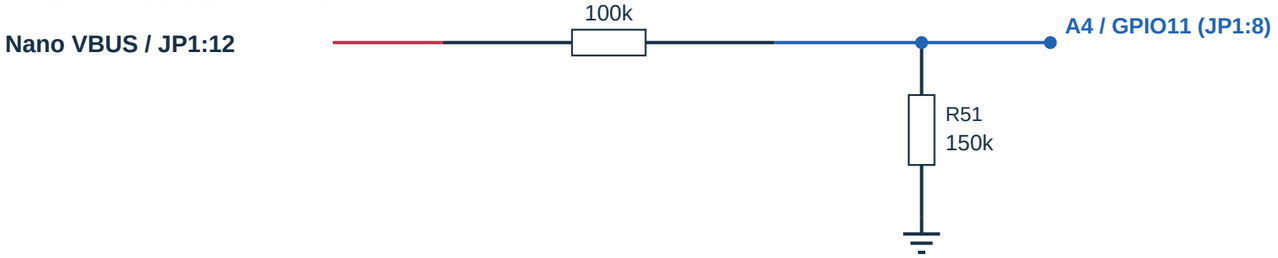
External power



C1: place close to VIN/GND; electrolytic positive terminal goes to +9V.

U1 external supply terminals:
VIN (JP1:15) = +9V input
3V3 (JP1:2) = logic supply output
GND (JP1:14 / JP2:4) = common return

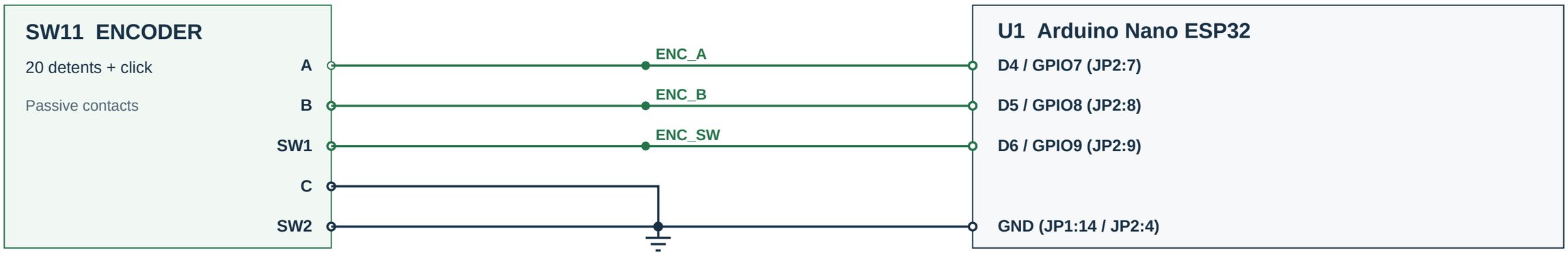
USB-C and VBUS sense



Native USB-C: MIDI / ROM serial flashing.
B1 to GND, tap RST, remove jumper to flash.
VBUS divider: 3.0V at 5.0V USB input.
Keep converter +5V separate from USB VBUS.
USB-only power leaves the LED rail off.

SW11 20-detent encoder with click - direct wiring

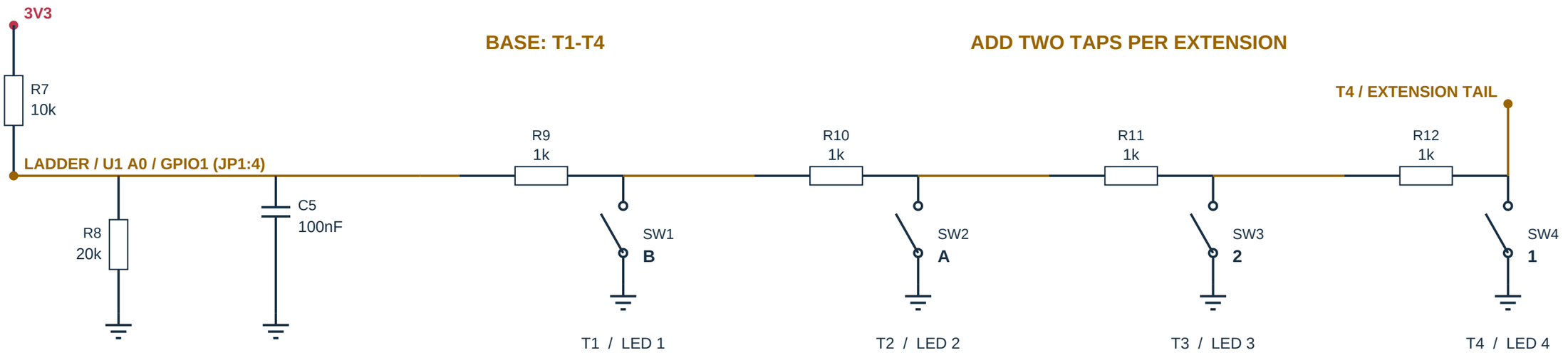
No external resistors or capacitors. Firmware enables internal pull-ups on GPIO7, GPIO8 and GPIO9.

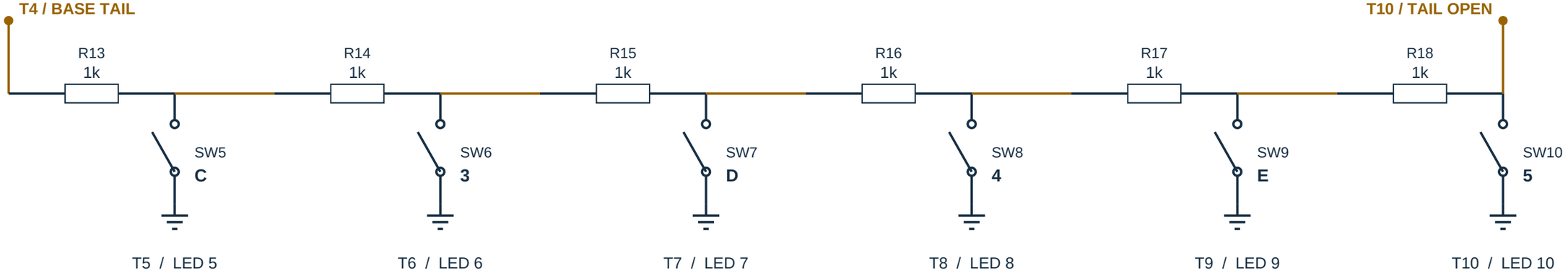


Functional pin arrangement: verify A/B/C and the two click terminals on the selected encoder.
Software decodes A/B and debounces the click. Keep the encoder harness short.

Modular footswitch resistor ladder

A single series-tap ladder supports 4, 6, 8 or 10 normally-open switches. Only one switch is detected at a time.





Modular harness and numbering

Each extension receives TAIL + GND and adds 2 x 1k.

Optional LED harness adds +5V and DOUT -> next DIN.

No terminating resistor at the final tail. All resistors 1%.

Select installed count in software, then capture calibration.

Left/right enclosure setting changes visual order only.

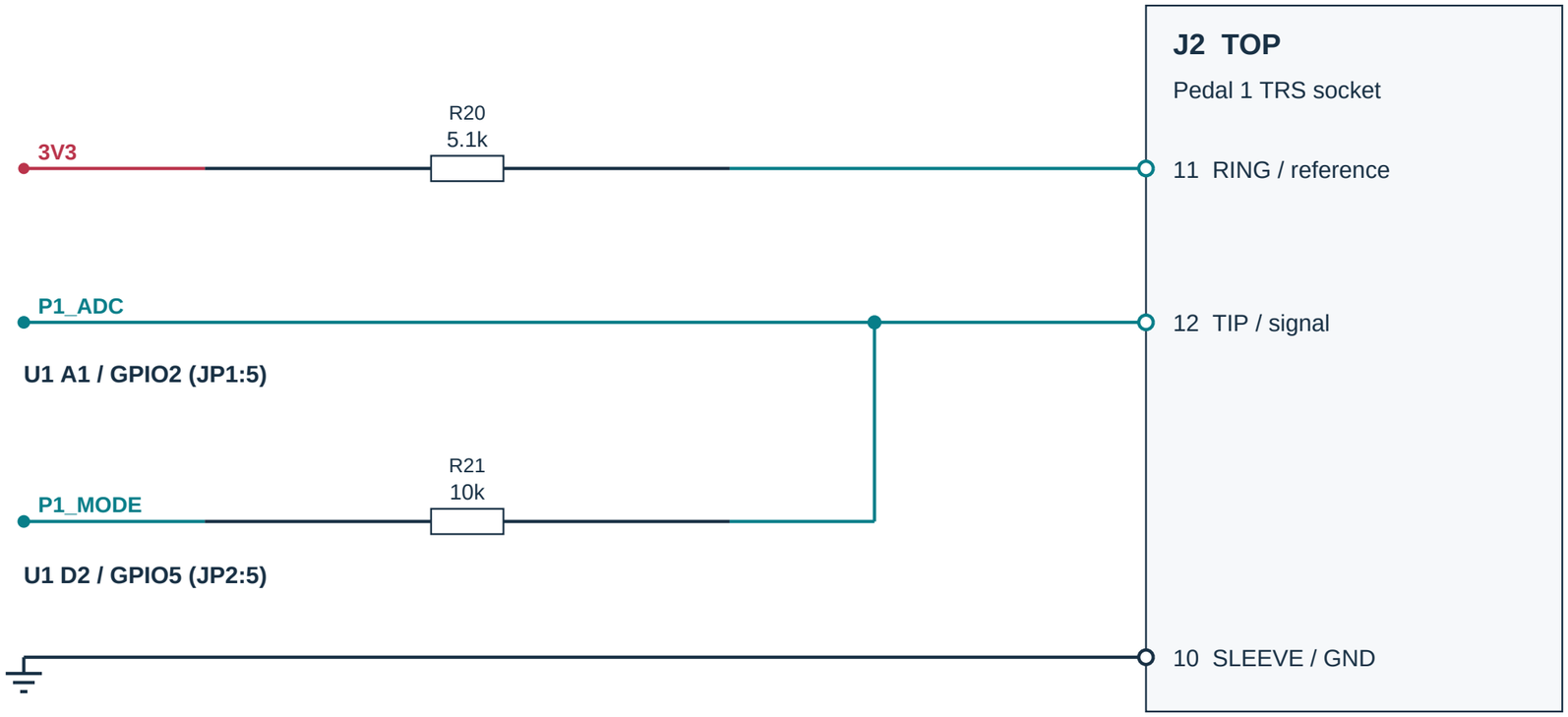
Tap	Name / LED	Expected mV	Tap	Name / LED	Expected mV
1	B / 1	287	6	3 / 6	1042
2	A / 2	508	7	D / 7	1127
3	2 / 3	683	8	4 / 8	1200
4	1 / 4	825	9	E / 9	1264
5	C / 5	943	10	5 / 10	1320

Idle: 2200mV at 3.300V. Values are nominal; calibrate the assembled ladder.

Continuous pedal input 1 - passive

Two resistors inside the controller. External pedal: passive variable resistor or potentiometer, 0-10k.

Inside the controller - only two resistors



Firmware bias state

Resistor mode: mode GPIO = HIGH.

Pot mode: mode GPIO = high impedance.

Disabled / boot: high impedance.

Internal pull-up and pull-down OFF.

Never drive the bias GPIO LOW.

Nominal values at 3.300V

10k variable resistor: TIP = 1650mV.

10k pot, full travel: TIP = 2185mV.

Ring short current: about 0.65mA.

ADC measures TIP directly (gain = 1).

Option A - three-wire potentiometer pedal



Capture actual heel and toe voltages after connecting.

Option B - two-wire variable resistor pedal



RING unused; a TS plug may connect it to SLEEVE.

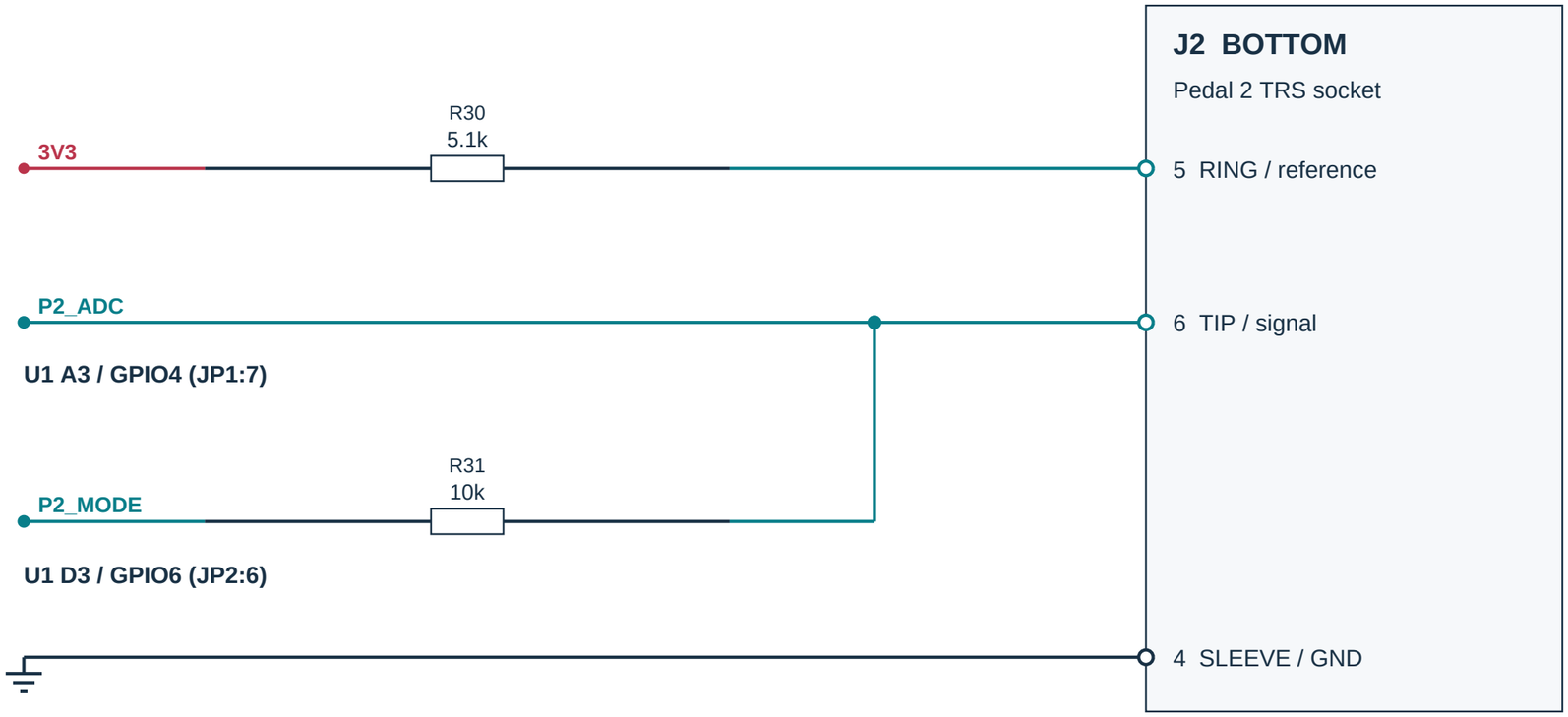
Select resistor mode and configure maximum resistance.

No active pedal interface, scaling divider or external filter capacitors. J2 switched contacts remain NC; pad13 mechanical.

Continuous pedal input 2 - passive

Two resistors inside the controller. External pedal: passive variable resistor or potentiometer, 0-10k.

Inside the controller - only two resistors



Firmware bias state

Resistor mode: mode GPIO = HIGH.

Pot mode: mode GPIO = high impedance.

Disabled / boot: high impedance.

Internal pull-up and pull-down OFF.

Never drive the bias GPIO LOW.

Nominal values at 3.300V

10k variable resistor: TIP = 1650mV.

10k pot, full travel: TIP = 2185mV.

Ring short current: about 0.65mA.

ADC measures TIP directly (gain = 1).

Option A - three-wire potentiometer pedal



Capture actual heel and toe voltages after connecting.

Option B - two-wire variable resistor pedal



RING unused; a TS plug may connect it to SLEEVE.

Select resistor mode and configure maximum resistance.

No active pedal interface, scaling divider or external filter capacitors. J2 switched contacts remain NC; pad13 mechanical.

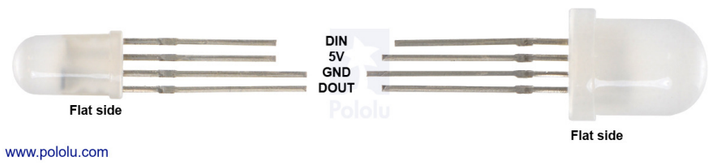
Pololu 2535 - direct LED chain

Only LEDs and wires: Nano D7 / GPIO10 drives the first DIN directly. All boxes use the manufacturer side-view lead order.

U1 Arduino Nano ESP32

- D7 / GPIO10 (JP2:10)
- GND (JP1:14 / JP2:4)

Pololu package orientation



Manufacturer side view: lens left, flat side down.

Direct wiring

D7 / GPIO10 -> LED1 DIN.

Each DOUT -> next LED DIN.

Every 5V lead -> regulated +5V.

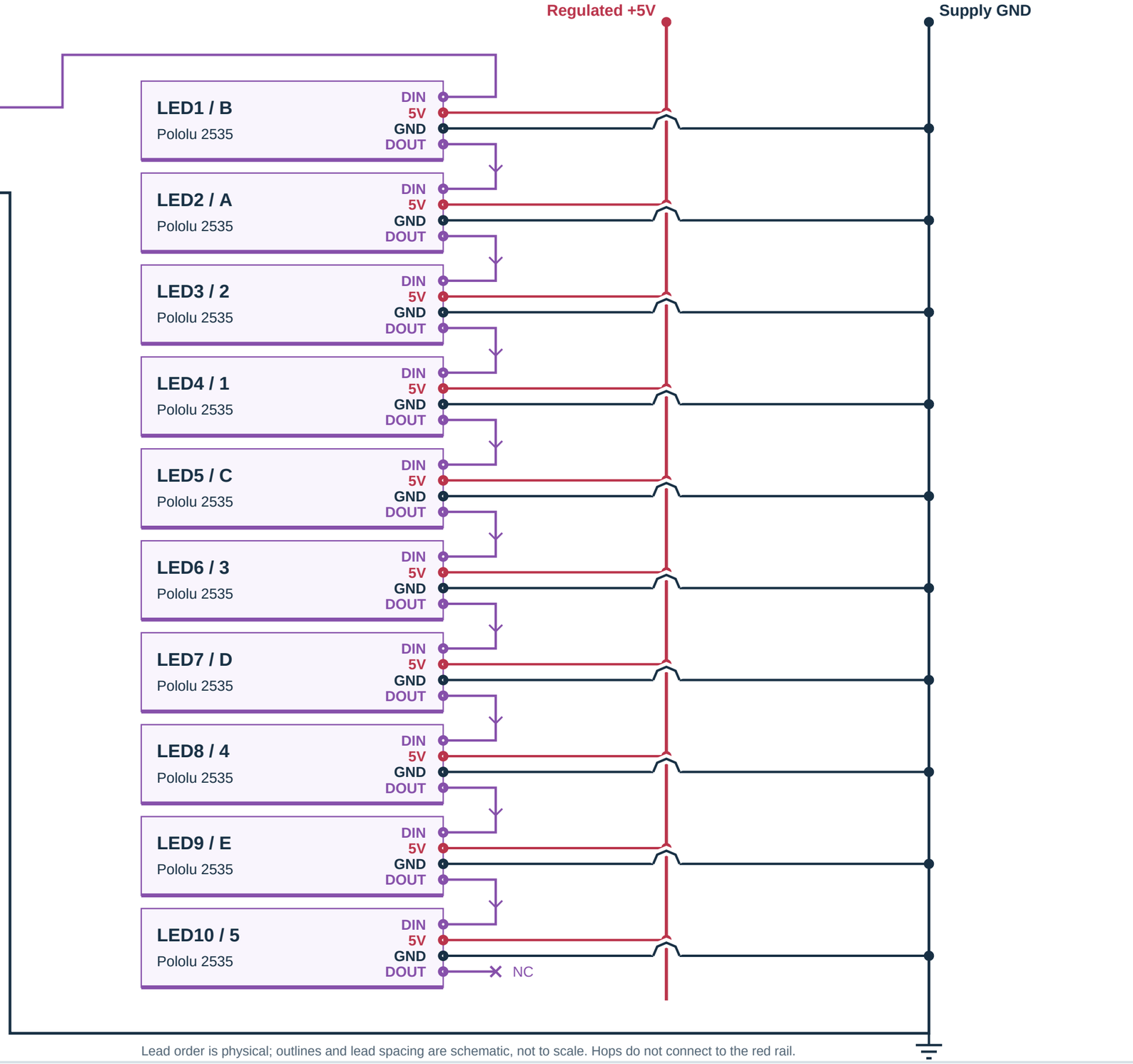
Every GND lead -> common GND.

Last DOUT -> leave unconnected.

RGB / 2535 / 800 kbit/s.

The driver is integrated into each LED.

Direct 3.3V GPIO drive at a 5V LED supply
requires verification with the actual LED batch.
No additional LED-chain components are fitted.

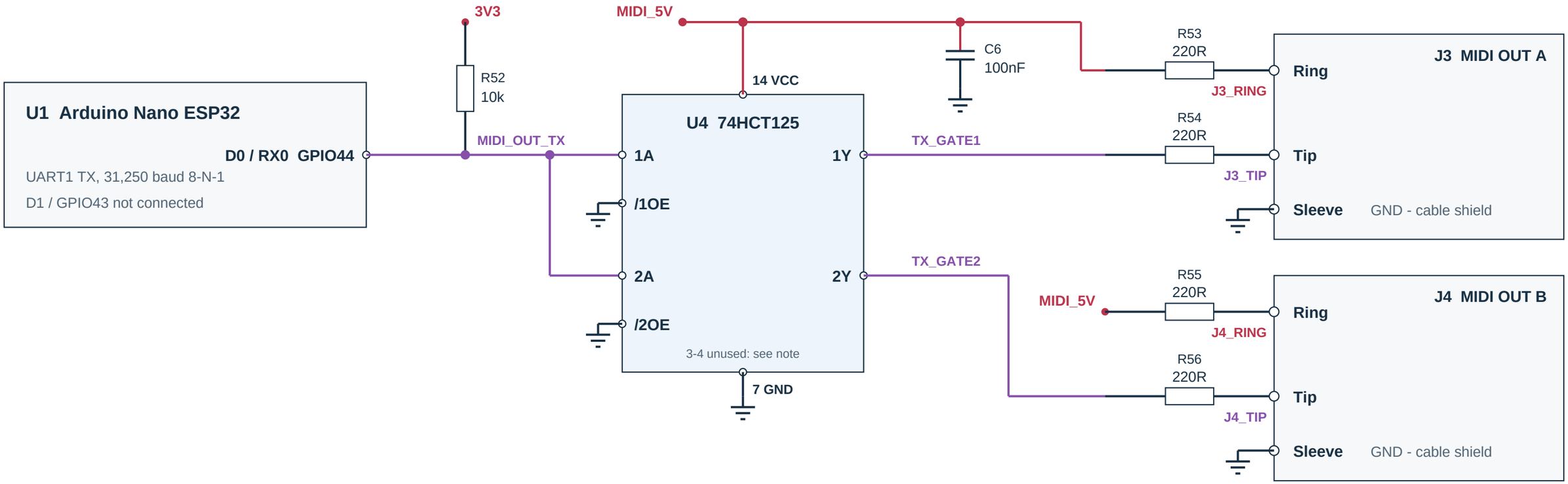


Lead order is physical; outlines and lead spacing are schematic, not to scale. Hops do not connect to the red rail.

MIDI OUT - two 3.5 mm TRS current loops

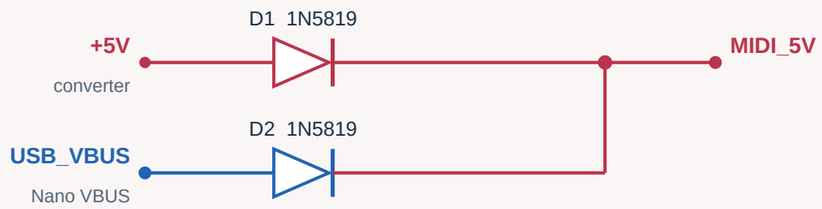
UART1 TX at 31,250 baud, 8-N-1; 74HCT125 gates 1 and 2 each drive one TRS Type A output. UART0 remains ROM-flash serial.

UART TX, idle pull-up and 74HCT125 buffer



J3 / J4 3.5 mm TRS, Type A

MIDI_5V supply / Schottky diode OR



Buffer and supply notes

R52 holds both buffer inputs idle-high during reset.
U4 /1OE and /2OE grounded; C6 directly across pins 14 / 7.
Unused inputs 9 / 12 and /OE 10 / 13 to GND;
outputs 8 / 11 left open.
Diodes isolate converter +5V and USB VBUS.

Each loop: MIDI_5V -> 220R -> ring -> receiver optocoupler -> tip -> 220R -> buffer output sink (J3: R53/R54/1Y, J4: R55/R56/2Y).

TRS Type A: tip = DIN pin 5, ring = DIN pin 4, sleeve = DIN pin 2. Each sleeve connects to GND to ground the cable shield.

One-way outputs only; no receive connection to the Nano. Use Type A adapters or cables for 5-pin DIN receivers.

Assembly reference, options and sources

The accompanying netlist and connection schedule include every external component terminal shown in this set.

Harness and assembly rules

All GND symbols are the same electrical net. Run analog returns separately to the controller ground point.

Keep LADDER and pedal signals away from LED power/data and the DC converter. Pair signals with ground.

Feed LED +5V/GND directly from the power distribution, rather than through the Nano headers.

The 10-pixel chain can draw about 0.5A at full white. Keep total load within the 1A converter rating.

LED lead order, lens pointing left and leads right, flat side down: top DIN, +5V, GND, bottom DOUT.

The LED chain contains only Pololu 2535 LEDs and wiring: GPIO10 directly to the first DIN.

Encoder terminal labels are functions; a specific encoder package pin numbering has not been selected.

GPIO numbers are logical; D/A labels and SCK/MOSI/TX names identify the external board pins.

Prototype verification

With power disconnected: continuity-check TRS terminals, power polarity, ground and all off-sheet nets.

Verify regulated +5V, then U1 3V3. Flash the passive-input firmware and recalibrate both pedals.

Check resistor-mode HIGH / pot-mode high impedance, ADC ranges, ladder taps, encoder, TRS, USB and BLE MIDI.

Each passive pedal input uses two resistors. Connect only unpowered variable-resistor or pot pedals.

This document does not establish ESD compliance, measured analog accuracy or simultaneous-supply qualification.

Source documents checked

Arduino Nano ESP32 ABX00083 pinout	Component-side top view, USB-C up; all 30 header pads
Arduino Nano ESP32 schematic V0.3	JP1: 1-15 top-down; JP2: 15-1 top-down
Waveshare 24382 external P2 connector	Local: design/fusion/datasheets/waveshare-24382-schematic.pdf
Amphenol ACJS-MHDC socket drawing	Local: design/fusion/datasheets/double-trs-ACJS-MHDC.pdf
Pololu S13V10F5 / #4083	5V, 1A step-up/down regulator; retailer item 1178565
Pololu #2535	5mm addressable RGB LED / WS2811 / 5V / 800kHz
Espressif self-powered USB devices	VBUS sense: 100k / 150k divider to A4 / GPIO11
MIDI Association - MIDI 1.0 electrical specifications	5 V current-loop interface; TRS Type A pin mapping

Source URLs, BOM, exact terminal connectivity and generated SVG sheets are in design/electrical/.